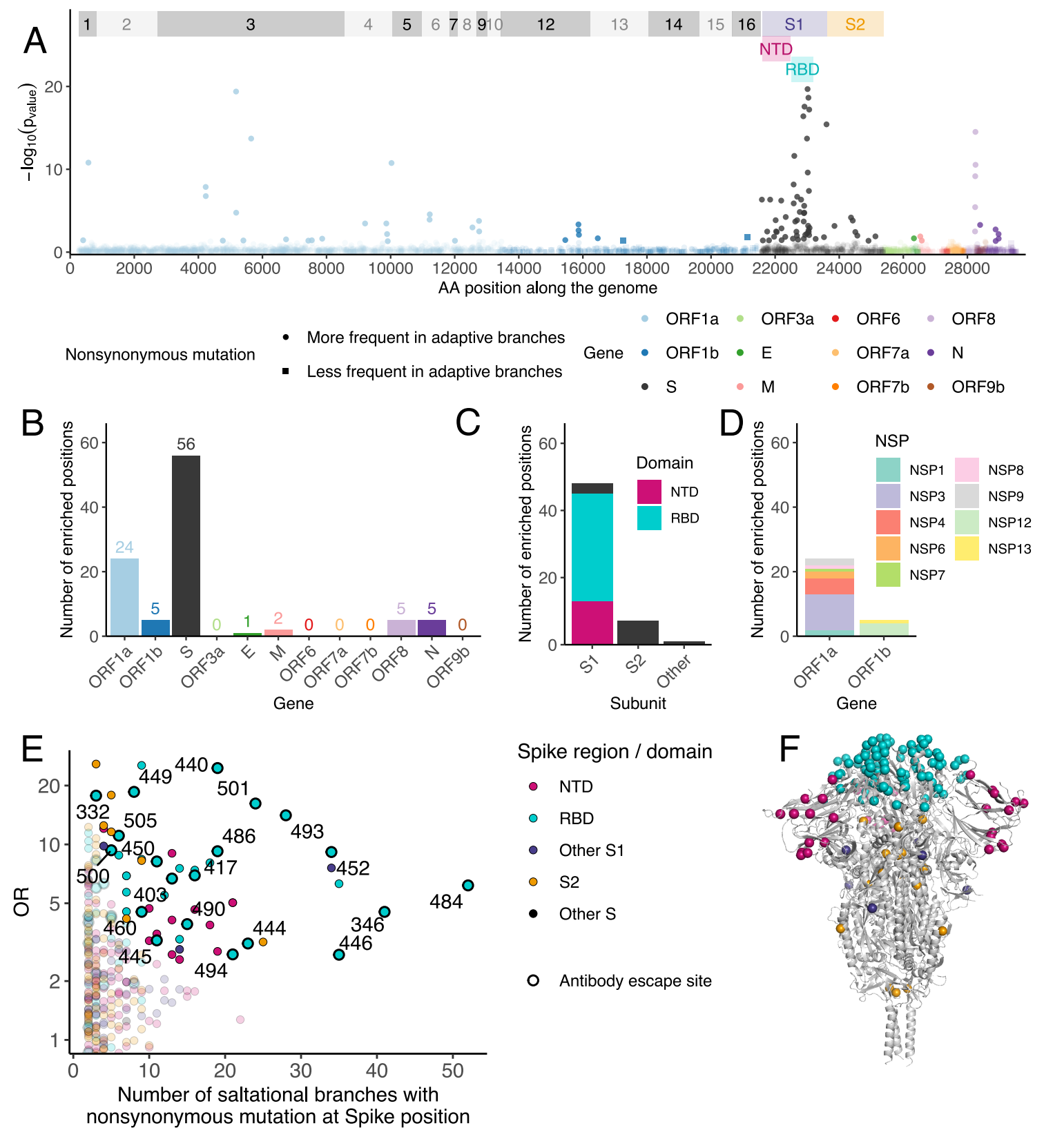

A
1 2 3 4 5 6 7 8 9 10 12 13 14 15 16 S1 S2
NTD
RBD
−log10(pvalue)
AA position along the genome
Nonsynonymous mutation
More frequent in adaptive branches
Less frequent in adaptive branches
Gene
ORF1a ORF1b S ORF3a E M ORF6 ORF7a ORF7b ORF8 N ORF9b
B
Number of enriched positions
Gene
C
Domain
NTD
RBD
Subunit
S1 S2 Other
D
NSP
NSP1 NSP3 NSP4 NSP6 NSP7 NSP8 NSP9 NSP12 NSP13
E
OR
Number of saltational branches with nonsynonymous mutation at Spike position
Spike region / domain
NTD
RBD
Other S1
S2
Other S
Antibody escape site
F